

Chapter 4 Exam

Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Approximate the number using a calculator. Round your answer to three decimal places.

1) $2^{3.1}$

1) _____

2) $2\sqrt{5}$

2) _____

Solve the problem.

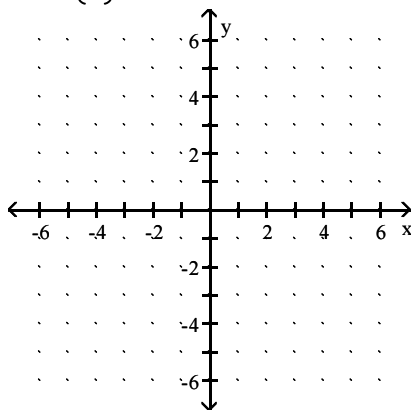
- 3) The rabbit population in a forest area grows at the rate of 6% monthly. If there are 170 rabbits in September, find how many rabbits (rounded to the nearest whole number) should be expected by next September. Use $y = 170(2.7)^{0.06t}$.

3) _____

Graph the function by making a table of coordinates.

4) $f(x) = \left(\frac{1}{3}\right)^x$

4) _____

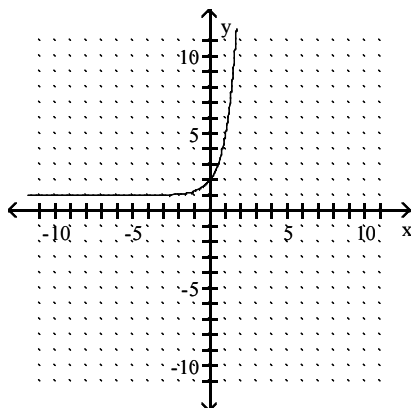


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

The graph of an exponential function is given. Select the function for the graph from the functions listed.

5)

5) _____



A) $f(x) = 4^x$

B) $f(x) = 4^x - 1$

C) $f(x) = 4^x + 1$

D) $f(x) = 4^x + 1$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Use the compound interest formulas $A = P\left(1 + \frac{r}{n}\right)^{nt}$ and $A = Pe^{rt}$ to solve.

- 6) Find the accumulated value of an investment of \$13,000 at 8% compounded semiannually for 10 years. 6) _____

Write the equation in its equivalent exponential form.

7) $\log_3 x = 2$ 7) _____

8) $\log_b 25 = 2$ 8) _____

Write the equation in its equivalent logarithmic form.

9) $5^3 = 125$ 9) _____

10) $3^{-3} = \frac{1}{27}$ 10) _____

11) $\sqrt[3]{216} = 6$ 11) _____

Evaluate the expression without using a calculator.

12) $\log_{10} 1$ 12) _____

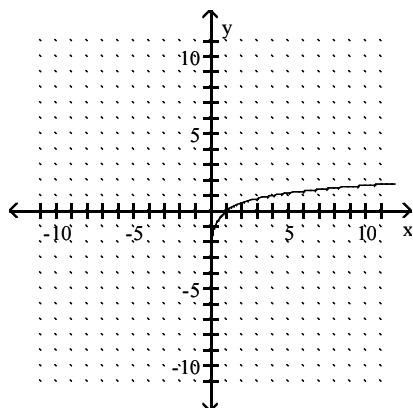
13) $\log_4 4$ 13) _____

14) $\log_9 9^{12}$ 14) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

The graph of a logarithmic function is given. Select the function for the graph from the options.

- 15) 15) _____



A) $f(x) = \log_4 (x + 2)$

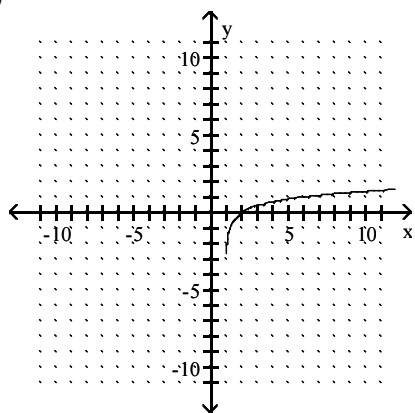
C) $f(x) = \log_4 x$

B) $f(x) = \log_4 x + 2$

D) $f(x) = \log_4 (x - 2)$

16)

16) _____



A) $f(x) = \log_5 x - 1$

B) $f(x) = \log_5 (x - 1)$

C) $f(x) = \log_5 x$

D) $f(x) = \log_5 (x + 1)$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the domain of the logarithmic function.

17) $f(x) = \log_3 (x + 6)$

17) _____

18) $f(x) = \ln (6 - x)$

18) _____

Use properties of logarithms to expand the logarithmic expression as much as possible. Where possible, evaluate logarithmic expressions without using a calculator.

19) $\log_3 (7 \cdot 11)$

19) _____

20) $\log_4 (16x)$

20) _____

21) $\log_3 \left(\frac{2}{5} \right)$

21) _____

22) $\log_3 3^{-5}$

22) _____

Use properties of logarithms to condense the logarithmic expression. Write the expression as a single logarithm whose coefficient is 1. Where possible, evaluate logarithmic expressions.

23) $2 \log_x 3 + \log_x 3$

23) _____

24) $\log_2 12 - \log_2 3$

24) _____

25) $\frac{1}{2} \log_6 x + \log_6 y$

25) _____

Answer Key

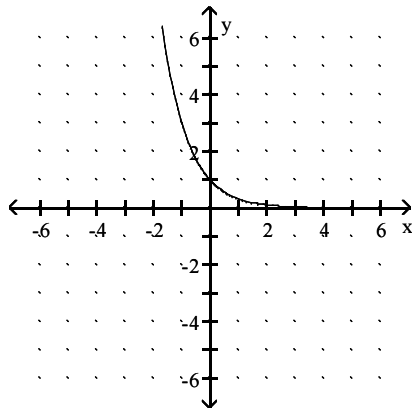
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1) 8.574

2) 4.711

3) 348

4)



5) D

6) \$28,484.60

7) $3^2 = x$

8) $b^2 = 25$

9) $\log_5 125 = 3$

10) $\log_3 \frac{1}{27} = -3$

11) $\log_{216} 6 = \frac{1}{3}$

12) 0

13) 1

14) 12

15) C

16) B

17) $(-6, \infty)$

18) $(-\infty, 6)$

19) $\log_3 7 + \log_3 11$

20) $2 + \log_4 x$

21) $\log_3 2 - \log_3 5$

22) $-5 \log_3 3$

23) $\log_x 27$

24) 2

25) $\log_6 y \sqrt{x}$

- 1) _____
- 2) _____
- 3) _____
- 4) _____

- 5) _____
- 6) _____
- 7) _____
- 8) _____
- 9) _____
- 10) _____

- 11) _____
- 12) _____
- 13) _____
- 14) _____
- 15) _____
- 16) _____
- 17) _____
- 18) _____
- 19) _____
- 20) _____
- 21) _____
- 22) _____
- 23) _____
- 24) _____
- 25) _____